

OBSERVATION TASK

Field Details

Course	Full Stack Web Development
Course Code	23CM4121
Experiment No.	Task1-2
Student Name	S.Deekshitha
Roll No.	A24126552117
Date	30-08-2026

1. TITLE

Explore CSS3 Styling and HTML5 Canvas Graphics

2. AIM

To understand how CSS3 enhances the appearance of web pages and to explore HTML5 graphics using the Canvas element.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic syntax and structure of CSS3.
- To apply CSS selectors, colors, fonts, borders, margins and padding.
- To style a webpage using backgrounds, gradients, box shadow, text shadow and rounded corners.
- To implement hover effects for navigation links and buttons.
- To draw basic shapes and text on a webpage using the HTML5 Canvas element.

4. THEORY

CSS3 (Cascading Style Sheets, Level 3) is the latest evolution of the CSS language used to control the presentation and layout of HTML documents. It introduces advanced styling features such as gradients, box shadows, text shadows, rounded corners (border-radius), and transition/hover effects, allowing developers to design visually appealing web pages without relying on images or scripts.

CSS rules follow the syntax `selector { property: value; }`. Selectors can target elements by tag name, class (.class), id (#id), or pseudo-classes such as `:hover`, which applies a style only while the mouse pointer is over an element.

HTML5 introduces the `<canvas>` element, a rectangular region on the page that can be used to draw graphics, shapes, and text dynamically using JavaScript. The 2D drawing context is obtained using `canvas.getContext("2d")`, which exposes methods such as `fillRect()` for rectangles, `arc()` for circles, `moveTo()/lineTo()/stroke()` for lines, and `fillText()` for text.

Together, CSS3 and HTML5 Canvas allow a single webpage to combine styled document content (header, navigation, cards, footer) with custom-drawn graphics, demonstrating both declarative styling and imperative, script-driven rendering.

5. ALGORITHM / PROCEDURE

- Create a new HTML file and set up the basic HTML5 document structure.

- Add a <style> block and define general/reset styles (margin, padding, box-sizing).
- Style the body with a font family and a linear gradient background.
- Create a header section and apply background color, text color and text-shadow.
- Create a navigation bar with links; style the links and add a :hover effect.
- Insert a <canvas> element inside a section and give it a border and background.
- Using JavaScript, get the 2D drawing context of the canvas.
- Draw a filled rectangle using fillRect().
- Draw a filled circle using beginPath(), arc() and fill().
- Draw a straight line using moveTo(), lineTo() and stroke().
- Display text on the canvas using fillText().
- Create three content cards with borders, border-radius, box-shadow and a button in each.
- Add a :hover effect on the buttons to change their background color.
- Create a footer with background color and centered text.
- Open the HTML file in a browser and verify the layout, styling and canvas drawing.
- Test hover effects on navigation links and buttons, and verify canvas shapes render correctly.

6. PROGRAM

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Styled Webpage</title>

<style>

/* Body */
body{
    margin:0;
    font-family:Verdana, sans-serif;
    background: linear-gradient(to top right, #ff7e5f, #feb47b);
}

/* Header */
header{
    background:linear-gradient(90deg, #ff7e5f, #feb47b);
    color:white;
    text-align:center;
    padding:25px;
}

header h1{
    text-shadow:3px 3px 5px gray;
}

/* Navigation */
nav{
    background:#333;
    overflow:hidden;
}

nav a{
    float:left;
    color:white;

```

```

nav a{
  float:left;
  color:■white;
  text-decoration:none;
  padding:15px 20px;
}

nav a:hover{
  background:■#ff9800;
  color:□black;
}

/* Main Section */
section{
  padding:20px;
}

/* Canvas */
canvas{
  display:block;
  margin:auto;
  border:3px solid □black;
  border-radius:10px;
  background:■white;
}

/* Cards */
.cards{
  display:flex;
  justify-content:space-around;
  margin-top:30px;
}

```

```

67
68 .card{
69   width:28%;
70   background:■white;
71   border:2px solid ■#ddd;
72   border-radius:15px;
73   padding:20px;
74   box-shadow:0px 4px 8px □gray;
75   text-align:center;
76 }
77
78 .card h2{
79   color:■darkblue;
80 }
81
82 .card p{
83   font-size:15px;
84 }
85
86 button{
87   background:■#2196F3;
88   color:■white;
89   border:none;
90   padding:10px 20px;
91   border-radius:20px;
92   cursor:pointer;
93 }
94
95 button:hover{
96   background:■#4CAF50;
97 }
98
99 /* Footer */
100 footer{

```

```

99  /* Footer */
100  footer{
101      margin-top:30px;
102      background: #222;
103      color: white;
104      text-align:center;
105      padding:15px;
106  }
107
108  </style>
109
110  </head>
111
112  <body>
113
114  <header>
115  <h1>My Styled Webpage</h1>
116  <p>HTML5 Canvas & CSS3 Demonstration</p>
117  </header>
118
119  <nav>
120  <a href="#">Home</a>
121  <a href="#">Gallery</a>
122  <a href="#">Services</a>
123  <a href="#">Contact</a>
124  </nav>
125
126  <section>
127
128  <h2 align="center">Canvas Drawing</h2>
129
130  <canvas id="canvas" width="600" height="300"></canvas>

```

```

130  <canvas id="canvas" width="600" height="300"></canvas>
131
132  <div class="cards">
133
134  <div class="card">
135  <h2>HTML5</h2>
136  <p>HTML5 provides semantic elements and multimedia support.</p>
137  <button>Know More</button>
138  </div>
139
140  <div class="card">
141  <h2>CSS3</h2>
142  <p>CSS3 improves the appearance of web pages with animations and effects.</p>
143  <button>Read More</button>
144  </div>
145
146  <div class="card">
147  <h2>Canvas</h2>
148  <p>Canvas is used to draw graphics like shapes, lines, text, and games.</p>
149  <button>Explore</button>
150  </div>
151
152  </div>
153
154  </section>
155
156  <footer>
157  <p>© 2026 My Styled Webpage</p>
158  </footer>
159
160  <script>
161
162  let canvas=document.getElementById("canvas");

```

```

<script>

let canvas=document.getElementById("canvas");
let c=canvas.getContext("2d");

/* Rectangle */
c.fillStyle="green";
c.fillRect(40,40,120,80);

/* Circle */
c.beginPath();
c.arc(250,80,40,0,2*Math.PI);
c.fillStyle="orange";
c.fill();

/* Line */
c.beginPath();
c.moveTo(350,40);
c.lineTo(550,140);
c.strokeStyle="red";
c.lineWidth=4;
c.stroke();

/* Name */
c.font="28px Georgia";
c.fillStyle="blue";
c.fillText("Deekshitha",180,230);

</script>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
linear-gradient()	Creates a smooth transition between two or more colors, used as the page background.
text-shadow	Adds a shadow behind text; used on the header title for depth.
:hover	Pseudo-class that applies styles only while the mouse pointer is over the element (nav links, buttons).
border-radius	Rounds the corners of an element, used on cards and buttons.
box-shadow	Adds a drop shadow around an element, used on the content cards.
getContext("2d")	Returns the 2D rendering context of the canvas, used to draw shapes and text.
fillRect()	Draws a filled rectangle at a given position and size on the canvas.
arc()	Draws a circular arc (used with fill() to draw a filled circle).
moveTo() / lineTo() / stroke()	Defines and draws a straight line path on the canvas.
fillText()	Renders text at a specified position on the canvas.

8. TEST CASES AND EXECUTION DATA

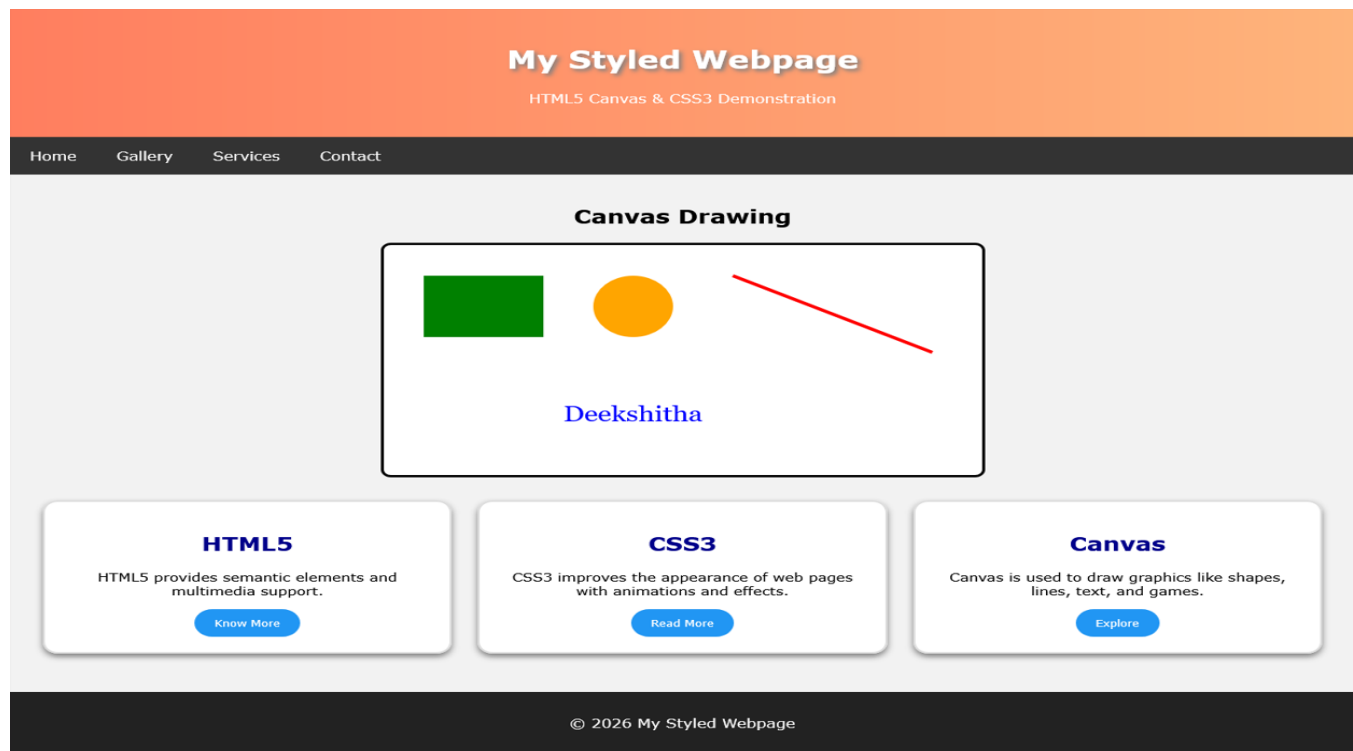
Each styling feature and canvas element was verified individually in the browser.

Test Case	Feature Tested	Expected Result	Actual Result	Status
TC-01	Page load	Header, nav, canvas, cards and footer render correctly	Rendered correctly	Pass
TC-02	Gradient background	Body shows a smooth color gradient	Gradient displayed correctly	Pass
TC-03	Canvas shapes	Rectangle, circle, line and text appear on canvas	All shapes and text displayed	Pass
TC-04	Nav link hover	Link background turns white with rounded corners on hover	Hover effect worked	Pass
TC-05	Button hover	Button background changes to green on hover	Hover effect worked	Pass
TC-06	Card styling	Cards show border-radius and box-shadow	Displayed correctly	Pass

9. OUTPUT

Output 1: Full Page View

The page displays a styled header with a shadowed title, a navigation bar, the HTML5 Canvas drawing (rectangle, circle, line and name text), three styled cards with buttons, and a footer, all rendered in the browser as shown below.



10. RESULT

Thus, a webpage titled "My Styled Webpage" was successfully designed using CSS3 styling techniques (gradients, shadows, borders, rounded corners and hover effects) and HTML5 Canvas graphics (rectangle, circle, line and text). All components were verified in the browser and the required visual effects were achieved as expected.